National Institute Of Technology Calicut
Department of Computer Science and Engineering
CS4043D Image Processing
Assignment 2

Date of Submission : On or before 20-04-2026
Maximum Marks : 15

Instructions to the students:

- *Any submitted work should be your own. Academic dishonesty in any form can lead to zero marks for the assignment.*
- *Any work submitted after the submission deadline will not be considered for evaluation (exception may be given only for genuine reasons).*
- *Students are free to use the language/tool of their choice.(Preferably Matlab/Octave/Java/Python etc.)*
- *Prepare a document that contains the question, your code and output. Save it with file name as Your Name_Rollno and upload it in the submission link given in eduserver on or before the deadline.*

1. On sample artificial, 8X8 image apply the following image transforms and print the image and the transform side by side.
   a) DFT b) DCT c) Walsh Transform
2. Take a real image(Your own photograph of size 256x256) and do segmentation using the following approaches
   a) Apply automatic global thresholding.
   b) Divide the image into 8x8 blocks and apply local thresholding to each block.
   c) Region Growing
3. Apply Runlength encoding(RLE) to the image obtained in question 2.a(Global thresholding). And print the result and the file size after applying RLE. Print the compression ratio.